Fitted Models versus Simple Rules: What Logistic Regression Reveals about Cross-Sectional Return Prediction in Indian Equities

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Abstract

I fitted unregularised logistic classifiers on five within-date standardised characteristics of Nifty 500 constituents (1,813,407 stock-dates, 2011–2026). They predicted four within-date rank events over 20 trading days, on twelve rolling folds (three-year training, one-year test, 21-day purge). Over test years 2014–2025, the classifier for beating the median had out-of-sample skill of 0.0027 and AUC of 0.537. Relative-volatility classification was much stronger (skill 0.129, AUC 0.758). A post hoc, approximate check suggests that ranking stocks by current volatility alone does about as well, so this mostly reflects persistence. In monthly portfolios, the two "avoid" models largely reproduced the simple exclusion of the most volatile fifth. The "outperform" model trailed a momentum-minus-volatility comparator (Sharpe difference −0.119; 95% CI −0.244 to −0.040), but that comparator was designed after the results were seen. All results are gross of costs and include 2022–2025, the period the wider project had reserved as a hold-out.

Read before interpreting results.

- **Sample.** Walk-forward test years 2014–2025 (training data 2011–2024). Each fold's model is out-of-sample for its test year, but no result is a project-level hold-out result: the 2022–2025 test years, training windows containing 2022–2024, and two later design choices (the quadratic high-volatility model and the momentum-minus-volatility comparator) used data the wider project had reserved as its hold-out from 2022-01-01.
- **Multiple testing.** Ledger family `SBX`: 15 pre-committed verdicts (`SBX-2.1`–`2.8`: four events × two classifier rules; `SBX-3a.1`–`3a.4`: four portfolios versus baseline; `SBX-3b.1`–`3b.3`: three model-versus-comparator pairs). The family is recorded separately from the legacy 32-test ledger. Rules were stated in each notebook's header cell before its results cells; no separate time-stamped registration exists. Everything else here is a labelled post hoc diagnostic.
- **Costs.** All portfolio returns are gross. An illustrative cost line (one-way monthly turnover $\times 2 \times 12\times$ an assumed 0.2% per side) is shown but not deducted. Spreads, impact and capacity were not modelled.
- **Point-in-time universe.** The universe is a snapshot-based point-in-time approximation of Nifty 500 membership. Semi-annual snapshots are carried forward to a daily flag, and a card requires `in_nifty500_flag == 1`. The 2018-06-30 snapshot is truncated, so up to ∼22% of the July–December 2018 universe may be misclassified. Stocks are equal-weighted, so small, less liquid members carry the same weight as large caps.
- **Other.** The Phase 2 classifier intervals come from the original resampling of calendar months i.i.d. ($B = 1000$). A moving-block rerun (2-, 3- and 6-month blocks) has been specified but not yet run, so the two 95% lower bounds closest to their thresholds (outperform and top-performer skill) may not survive it. Classification labels are within-date ranks, but portfolio returns are absolute and long-only, and market betas were not estimated.

1 Introduction

Fitted classifiers are a standard first tool for cross-sectional stock prediction, and they always produce a score. A harder question is whether the fitted weights do better than the simplest rule built from the same characteristics. This paper asks that question for five well-known characteristics on Nifty 500 constituents: momentum, short-term reversal, realised volatility, turnover and overnight drift.

A fitted model has two possible uses. As a *selection rule*, its scores are what you trade. As a *diagnostic*, its coefficients and the portfolios it picks show which characteristics carry information. The two uses can disagree. A model can identify the right characteristics but weight them worse than a fixed rule, through estimation noise or a mismatch between the training objective and the portfolio's use of the scores. This is the out-of-sample pattern behind the finding that equal-weight or "improper" linear rules often match estimated weights [3, 4].

Answer in brief. Across the tested folds, the five-characteristic logistic models contained little information about which stocks would beat the cross-sectional median: skill 0.0027 (log-loss reduction relative to the base rate) and per-date AUC 0.537. Relative future volatility was far more predictable (skill 0.129, AUC 0.758). A post hoc check suggests that most of this reflects persistence in volatility rather than anything the fitted model adds. The two "avoid" portfolios held 95.8% and 96.9% of the same stocks as an unfitted exclusion of the most volatile fifth. Their Sharpe-ratio advantage over that rule was small. It was marginal at 95% for the crash model and not robust to the three-comparison interval. The "outperform" portfolio trailed a fixed momentum-minus-volatility comparator, but that comparator is a retrospective benchmark designed after the results were seen. The findings are diagnostic, not prospective evidence of a tradable strategy.

The companion work [6] studies regime-dependent momentum and reversal information coefficients on the same universe. Related working papers examine momentum [8], path-texture measures [9] and mutual information [7]. This paper does not test regimes. It asks only whether fitting adds value beyond fixed sorts.

2 Data and universe

Panel and universe. Inputs come from a frozen daily feature panel (`feature_panel_with_pit_v2_2026-07-24`; 1,069 tickers, 4,237 dates), whose construction is documented in the `data_pipeline` folder of the code repository. A stock-date becomes a *card* only if `in_nifty500_flag == 1` on that date. Feature windows may look back into dates when the stock was outside the index.

Cards. All windows are counted in each ticker's own trading rows, with day t the card date. The five features use information up to the close of t:

- momentum 12–1: daily returns compounded over rows $t-252$ to $t-22$;
- reversal: the 21 daily returns in rows $t-21$ to $t-1$ (the builder's "prior 21 returns" convention, which excludes day t's own return);
- realised volatility: the panel's 60-day column at t;
- log-turnover: the log of day-t turnover;
- overnight drift: the mean of the overnight returns in rows $t-21$ to $t-1$.

Missing returns, including the vendor-failure date 2025-03-18, are dropped from products, never zero-filled. The target is the *absolute* return from the open of $t+1$ to the open of $t+21$. The builder checked on a sample card that the feature cutoff precedes target entry, and it recomputed momentum and the target for two cards from raw prices (difference 0.0). After dropping cards with any missing feature ($-65{,}678$) or a missing target ($-1{,}982$), and dates with fewer than 30

names (−350 cards on 67 dates), the dataset contains 1,813,407 cards on 1,041 tickers and 3,756 dates (2011-01-03 to 2026-06-10). Each feature is z-scored within each date after filtering.

Known data defects. Two defects are disclosed here. Neither is corrected in this version.

1. *The 2025 volatility hole.* The dataset has no cards from 2025-03-17 to 2025-06-17 (62 trading days). One missing daily return on 2025-03-18 blanks the panel's 60-day volatility for about 60 sessions. Inside that window, 97.4% of in-index rows lack it, compared with 0.3% before. The builder drops cards with a missing feature, so the portfolios sit out April–June 2025 (141 invested months instead of 144). The issue affects any analysis that uses this column and reaches into 2025. The fix, a NaN-tolerant rebuild, belongs in the panel repository.
2. *Exchange-closed dates.* The panel's date index contains some exchange-closed dates on which most tickers are blank. This study inherits that panel. The 67 dates removed for having fewer than 30 names were not itemised, so I cannot say how many were closed dates. A closed-date row inside a ticker's history would also be counted as a trading row by the positional windows. The Phase 3 rebalance plan found no late or skipped month outside April–June 2025. A calendar-corrected rebuild of the panel is planned. Its effect on the numbers here is unknown but expected to be small.

Hold-out status. "Out-of-sample" means two different things here. At the *fold* level, each test year was scored by a model fitted only on earlier data, with a purge. At the *project* level, the wider project had reserved 2022-01-01 onward as an untouched hold-out, and this study did not preserve it. The folds testing 2022–2025 were fitted on windows that include 2022–2024 data, their test scores enter every headline number, and two later choices used 2014–2025 results. The sandbox's build log describes the work as having "no hold-out spent". That phrase means no formal project-level hold-out protocol was followed; it does not mean 2022+ data were unused. For these five features, a clean check now requires forward tracking under a frozen protocol.

3 Method

3.1 Feature geometry (descriptive)

The averages over all dates of the within-date correlations between the z-scored features (dates with at least 30 names) were all small in magnitude. The largest was reversal versus overnight drift at +0.217; momentum versus realised volatility was +0.016. These are averages over dates; their dispersion across dates was not saved, so I do not claim the features are nearly uncorrelated on every date. As an illustration only, a principal-component decomposition written from scratch in numpy and run on one date (2017-07-04) gave variance shares of 28.4, 22.6, 19.4, 17.9 and 11.6%. These matched scikit-learn to 1.1×10^{-16}. On three other inspected dates, the first two components explained 52–59%.

3.2 Events and models (Phase 2)

Four binary events were defined from within-date percentile ranks:

- *outperform*: forward return above that date's median (base rate 0.50);
- *top performer*: forward return in the top 20%;
- *crash*: forward return in the bottom 20%;
- *high volatility*: forward realised volatility in the top 20%. This is the standard deviation of close-to-close daily returns over rows $t + 1$ to $t + 20$, with at least 10 observations. It is deliberately a close-to-close window, unlike the open-to-open return target.

Every label is a within-date rank and every feature a within-date z-score, so the classifiers target relative outcomes and cannot gain by forecasting market-wide moves. Three nested

models were fitted for each event: M0 (intercept only), M1 (five linear terms) and M2 (M1 plus the five squared terms). Fitting used unregularised maximum likelihood (scikit-learn `LogisticRegression`, no penalty, L-BFGS, up to 5,000 iterations). On one fold, a from-scratch gradient-descent fit reproduced the coefficients to within 8.5×10^{-5}.

3.3 Validation scheme and metrics

The validation was walk-forward. Each of 12 folds trained on three calendar years and tested on the next (test years 2014–2025). The last 21 trading dates before each training cutoff were purged. An audit confirmed for every fold that the last training card's exit date fell before 1 January of the test year. There was no tuning and no embargo beyond the purge.

The two metrics weight observations differently:

- *Skill* $= 1 - \mathrm{LL}(\text{model})/\mathrm{LL}(\text{M0})$, with LL the log-loss *summed over all test cards*. Dates with more names therefore carry more weight.
- *AUC* is computed *per date* and then averaged with equal weight on each date.

Pre-committed rules: (1) M1 beats M0 if the skill CI lies above 0, the AUC CI lies above 0.5, and skill is positive in both halves (test years 2014–19 and 2020–25); (2) M2 beats M1 if the gain CI lies above 0, the gain exceeds 0.002 (a materiality floor), and the gain is positive in both halves. A suspicion rule required a leakage hunt if a return-event AUC exceeded 0.60, or 0.65 for crash. Intervals are 95% percentiles from resampling calendar months i.i.d. ($B = 1000$, seed 0). This is a block bootstrap with one-month blocks, so cards whose forward windows cross a month boundary leave some dependence between blocks.

3.4 Portfolios (Phase 3a) and comparators (Phase 3b)

On the first trading day of each month, the in-index stocks were ranked by their out-of-sample predicted probability. Only the ranking was used, so calibration plays no role in portfolio selection. With n stocks and $k = \mathrm{round}(0.2n)$, "pick" portfolios hold the top k and "avoid" portfolios hold the other $n - k$. Stocks are equal-weighted, bought at the open of $t + 1$ and sold at the open of $t + 21$. The baseline holds every stock. The portfolios are: pick outperformers (M1 outperform), pick big winners (M1 top), avoid crashes (M1 crash) and avoid wild movers (M2 high volatility, the only event that passed rule 2).

Phase 3b added unfitted comparators ("twins") built from the same date-t z-scores:

- *cut most volatile*: hold all stocks except the 20% with the highest z_{vol}; this is the comparator for both avoid portfolios;
- *momentum minus volatility*: hold the 20% with the highest $z_{\mathrm{mom}} - z_{\mathrm{vol}}$; this is the comparator for pick outperformers;
- *momentum only*: hold the top 20% by z_{mom}; this is a reference, not a verdict.

The comparators were specified after Phase 3a showed which characteristics the model portfolios tilted toward. They are *retrospective, hindsight-informed benchmarks*, not untouched challengers.

The performance measures were defined as follows:

- the Sharpe ratio is the raw monthly mean return divided by the monthly standard deviation (ddof 1), multiplied by $\sqrt{12}$, with no risk-free rate subtracted;
- one-way turnover is $\frac{1}{2}\sum_i |w_{i,t} - w_{i,t-1}|$ between consecutive target portfolios, averaged over months;
- maximum drawdown is measured on the compounded monthly wealth curve.

Differences were tested with a paired moving-block bootstrap of monthly returns: the same months for both portfolios, 6-month blocks, $B = 2000$, seed 0, percentile intervals. Two rules were pre-committed:

- *Phase 3a:* an avoid portfolio "reduces risk" if the volatility-difference CI is below 0 and its drawdown is smaller than the baseline's, and it is "worth it" if the Sharpe-difference CI is

Table 1: Out-of-sample classifier scorecard, test years 2014–2025 (includes 2022–2025). Skill is the log-loss reduction relative to M0, summed over all test cards; AUC is the average of per-date AUCs. Intervals are 95% percentiles from i.i.d. resampling of calendar months ($B = 1000$), as originally computed; the moving-block rerun is pending. R1: M1 beats M0. R2: M2 materially beats M1.

Event	M1 skill [95% CI]	M1 AUC [95% CI]	Skill 14–19/20–25	M2−M1 skill [95% CI]	R1	R2
Outperform	0.0027 [0.0002, 0.0052]	0.537 [0.526, 0.549]	0.0050 / 0.0003	−0.0001 [−0.0004, 0.0002]	yes	no
Top performer	0.0022 [0.0006, 0.0037]	0.537 [0.527, 0.546]	0.0021 / 0.0022	0.0003 [−0.0003, 0.0009]	yes	no
Crash	0.0192 [0.0134, 0.0245]	0.596 [0.582, 0.609]	0.0282 / 0.0098	0.0013 [0.0005, 0.0021]	yes	no
High vol	0.1292 [0.1205, 0.1378]	0.758 [0.751, 0.765]	0.1550 / 0.1025	0.0143 [0.0117, 0.0170]	yes	yes

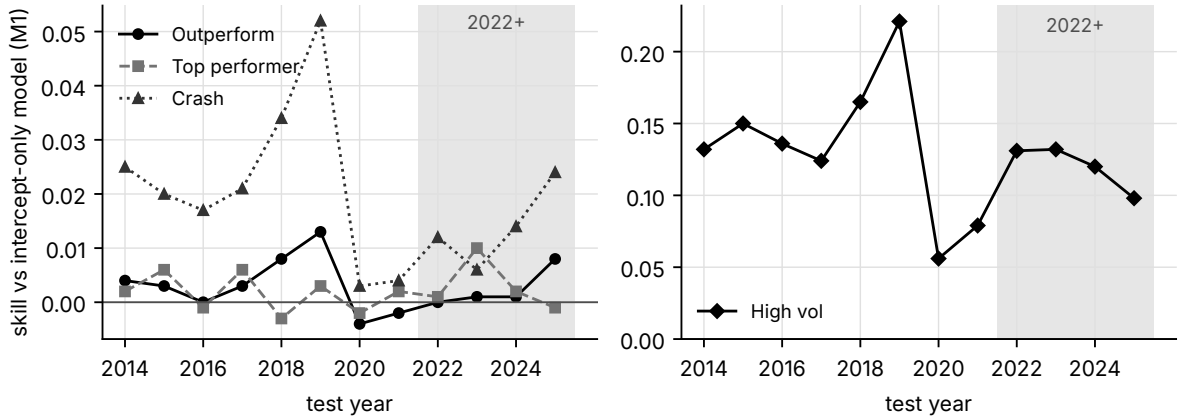


Figure 1: Out-of-sample M1 skill by test year. Left: the three return events. Right: high volatility (different scale). The shaded years 2022–2025 lie in the period the wider project had reserved as a hold-out. Values are parsed from the Phase 2 notebook output (rounded to 3 decimals); no per-year intervals were computed.

above 0. A pick portfolio "adds return" if the active-return CI is above 0 and active return is positive in both halves.

- *Phase 3b:* on the Sharpe difference, model minus comparator, a 95% CI entirely above 0 means the model adds value, entirely below 0 means the comparator is better, and otherwise there is no detectable difference.

The 98.75% intervals in Phase 3a correspond to its four comparisons with the baseline, and the 98.33% intervals in Phase 3b to its three pairs.

4 Results

4.1 What the classifiers could predict

Table 1 reports the scorecard. Under the original intervals, M1 passed rule 1 for all four events, but the effect sizes differed by a factor of about 50. The outperform classifier's skill was 0.0027 overall and only 0.0003 in 2020–25, and its 95% lower bound (0.0002) is the one most exposed to the pending block rerun. Curvature (M2) passed the materiality rule only for high volatility. For crash, the gain CI excluded zero, but the gain of 0.0013 fell below the 0.002 floor. No return-event AUC approached its suspicion threshold (best 0.537, 0.540 and 0.598 against limits of 0.60, 0.60 and 0.65). The largest calibration gap over ten probability bins was 0.050 (high volatility, M1), and the other models' gaps were at most 0.036. Figure 1 shows skill by test year.

How much of the volatility result is persistence? (post hoc). A model-free analysis on the same dataset reports the rate of the identical high-volatility event in each decile of current

Table 2: M1 odds ratio per one cross-sectional standard deviation: median across the 12 folds, with the number of folds in which the odds ratio exceeded 1 in brackets. Test years 2014–2025.

Event	Momentum	Reversal	Realised vol	Log-turnover	Overnight drift
Outperform	1.094 [12]	1.002 [6]	0.892 [0]	0.994 [5]	0.975 [0]
Top performer	1.095 [12]	1.006 [6]	1.110 [12]	0.970 [4]	0.992 [5]
Crash	0.897 [0]	0.999 [6]	1.388 [12]	1.031 [8]	1.043 [12]
High vol	0.912 [3]	0.936 [3]	2.597 [12]	0.920 [1]	1.093 [12]

Table 3: Monthly-rebalanced equal-weight portfolios: 141 invested months, January 2014 – December 2025 (April–June 2025 sat out). All figures are gross of costs. Sharpe is the monthly mean over the monthly SD $\times\sqrt{12}$, with no risk-free rate. Turnover is one-way per month. The last column, turnover $\times 2 \times 12 \times 0.2\%$, is illustrative and not deducted.

Portfolio	CAGR	Vol	Sharpe	Max DD	Turnover	Cost line/yr
Baseline (all stocks)	12.6%	21.3%	0.668	−45.4%	0.017	0.1%
Pick outperformers (model)	17.9%	16.3%	1.100	−26.4%	0.350	1.7%
Pick big winners (model)	11.4%	23.3%	0.586	−52.4%	0.334	–
Avoid crashes (model)	15.2%	19.8%	0.822	−36.1%	0.085	0.4%
Avoid wild movers (model)	14.9%	19.7%	0.808	−37.5%	0.080	0.4%
Comparator: cut most volatile	14.8%	20.0%	0.797	−37.0%	0.080	0.4%
Comparator: momentum − volatility	21.3%	17.2%	1.219	−25.2%	0.286	1.4%
Reference: momentum only	19.6%	20.7%	0.978	−29.8%	0.226	1.1%

realised volatility. The decile edges came from the trailing 252 days of z-scores, pooled over 2012-01-07 to 2026-06-10. The rate rose monotonically from 0.037 in the lowest decile to 0.537 in the highest. Treating the ten deciles as equal-sized implies an AUC of about 0.757 for ranking by current volatility alone, or 0.745 to 0.770 if the decile shares are allowed to vary by ±20%. That is essentially the classifier's 0.758 (M1) and 0.763 (M2). The comparison is approximate: it is pooled rather than averaged per date, uses a different sample span, and discards ordering within deciles, which should if anything understate the continuous ranking. It indicates that the high-volatility result mostly reflects the persistence of volatility. The fitted model adds little beyond it. An earlier sandbox analysis of monthly volatility states is consistent with this: a stock in the top volatility quintile in one month was in it the next month with probability 0.426, against a 0.20 base rate. An exact per-date comparison is part of the pending rerun.

Table 2 shows the coefficients as a diagnostic. Realised volatility had the largest and most stable coefficients: it raised the odds of high volatility (2.6× per SD) and of crash (1.39×) in all 12 folds. Momentum raised the odds of outperforming and of a top-quintile return in all 12 folds and lowered crash odds in all 12. Reversal changed sign across folds for every event. Overnight drift raised crash and high-volatility odds in all 12 folds. The top-performer model loaded positively on both momentum and volatility, which is consistent with volatility feeding both tails of the return distribution.

4.2 From scores to portfolios

Both avoid portfolios passed the pre-committed "reduces risk" and "worth it" rules (Table 3, Figure 2). Their volatility-difference CIs were $[-1.94, -1.15]$ and $[-1.96, -1.21]$ percentage points, and their Sharpe-difference CIs $[+0.085, +0.258]$ and $[+0.075, +0.237]$. Both Sharpe gains held at 98.75%. Neither pick portfolio passed "adds return". Pick outperformers earned +3.72%/yr of active return, with CI $[-1.60\%, +8.67\%]$ and halves of +6.0% and +1.3%. Its Sharpe advantage ($[+0.209, +0.726]$) came mainly from lower volatility (16.3% against 21.3%). These Sharpe gains come from long-only, absolute-return portfolios. A portfolio holding calmer

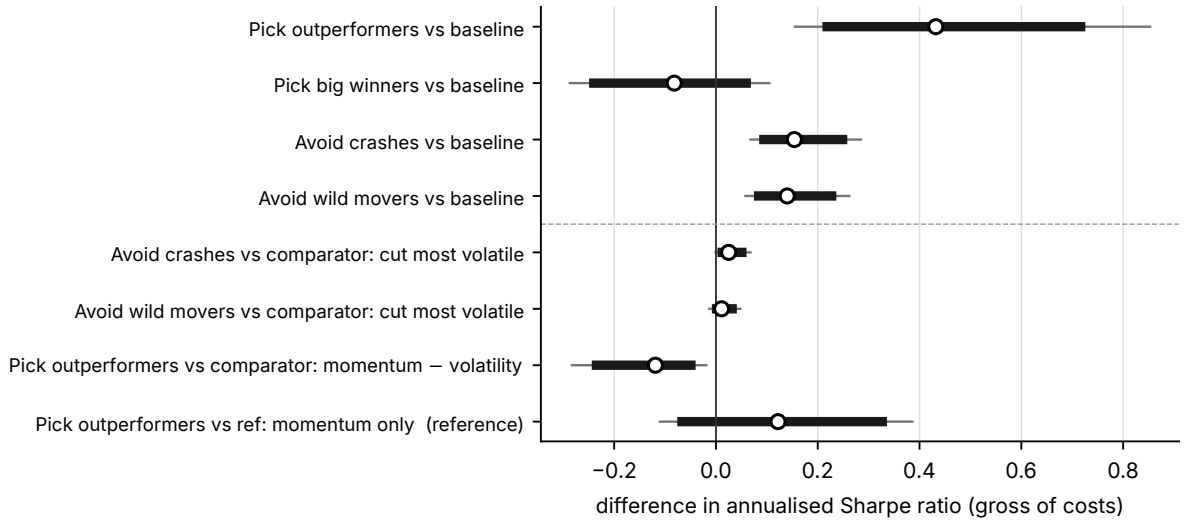


Figure 2: Differences in annualised Sharpe ratio, gross of costs, over 141 months (2014–2025, including the 2022–2025 hold-out period). Intervals come from a paired moving-block bootstrap (6-month blocks, $B = 2000$). Thick bars are 95% intervals. Thin bars are 98.75% above the dashed line (four comparisons with the baseline) and 98.33% below it (three pairs). Above the line: model portfolio minus baseline, with points equal to differences of the reported Sharpe ratios. Below the line: model minus the hindsight-designed comparator.

Table 4: Characteristics of the model portfolios: mean within-date z-score of held stocks over 141 months, and the mean (minimum) monthly share of the model's holdings that the comparator also held. Two random 80% subsets overlap about 80%; two random 20% subsets overlap about 20%.

Portfolio	Mom	Rev	Vol	Turn	Overn.	Overlap with comparator
Pick outperformers	+0.83	−0.01	−0.64	+0.19	−0.39	74.3% (48.0%) mom−vol
Pick big winners	+0.96	+0.07	+1.07	+0.11	+0.13	–
Avoid crashes	+0.06	−0.04	−0.36	−0.02	−0.11	95.8% (92.3%) cut-vol
Avoid wild movers	0.00	−0.02	−0.37	+0.01	−0.10	96.9% (93.2%) cut-vol

stocks can raise its Sharpe ratio through lower market exposure without better relative selection. Because betas and factor exposures were not estimated, the gains should not be read as stock-selection alpha. Both avoid portfolios' active returns fell from about 3%/yr in 2014–19 to under 0.6%/yr in 2020–25.

4.3 Model versus comparator

Table 4 shows why simple comparators were the natural benchmark. The stocks held by the avoid portfolios had mean volatility z-scores of −0.36 and −0.37, so the excluded fifth averaged about +1.45. Of the stocks excluded by avoid crashes, 86.2% (minimum 73.7%) were also excluded by avoid wild movers. The two prediction rankings had an average rank correlation of 0.93. Pick outperformers sat in the high-momentum, low-volatility corner.

The three pre-committed pairs (lower panel of Figure 2):

- **Avoid crashes vs. cut most volatile:** +0.025, 95% CI [+0.003, +0.060], 98.33% CI [−0.001, +0.068]. The model passes the unadjusted 95% rule, so the result is marginal at 95%, and it is not robust to the three-comparison interval. Active return was +0.33%/yr with CI [−0.13%, +0.91%].
- **Avoid wild movers vs. cut most volatile:** +0.011, 95% CI [−0.008, +0.041], so there is no detectable difference.

- **Pick outperformers vs. momentum − volatility:** −0.119, 95% CI [−0.244, −0.040], 98.33% CI [−0.283, −0.019], so under the pre-committed rule the comparator is better. Active return was −3.01%/yr with CI [−4.92%, −1.60%], negative in both halves. Because the comparator was designed after the model's tilts were observed, this shows only that, over this sample, a 1:1 combination of the two characteristics the model leaned on did at least as well as the fitted weights. It does not show that the comparator would beat the model going forward.

In the reference comparison with momentum only, the model's Sharpe was higher by +0.122 (95% CI [−0.076, +0.336]), which is inconclusive. All three comparators beat the baseline on Sharpe: cut most volatile [+0.071, +0.210], momentum − volatility [+0.306, +0.893] and momentum only [+0.073, +0.592]. Before the run, the notebooks expected the avoid pairs to show high overlap and little difference, which they did, and expected pick outperformers to tie its comparator, which it did not.

5 Robustness and diagnostics

Leakage and reproduction. Each prediction lies in its own fold's test year. A fresh fit of the 2019 crash model on pre-2019 rows reproduced the stored predictions exactly. Three forward returns recomputed from raw opens matched the dataset exactly. An independent recomputation matched the portfolio and comparator returns for two months to within 10^{-12}. Phase 3a reproduced Phase 2 skill to within 6×10^{-4}, and Phase 3b reproduced Phase 3a. On the M0 rung, intercepts equalled the logit of the training base rate and AUC was 0.500.

Universe and survivorship. None of the 68,670 candidate stock-months had a flag other than 1. For example, 63MOONS.NS was excluded after leaving the index on 2017-01-02. Twenty-three in-index stock-months (0.033%) had too little later data to compute a forward return; their median return to the last available price was +3.0%. Adjacent holding windows overlapped by 0.29 days on average (maximum 3), so monthly returns are close to non-overlapping.

Absolute versus relative target (verified diagnostics). Two earlier sandbox notebooks fitted ordinary least squares to the same five features on the same rolling scheme. One used the absolute target and one used the target minus that date's equal-weighted mean. I re-extracted their printed outputs for this revision. For the absolute target, the mean across the 12 folds of the per-fold out-of-sample R^2 was −0.0421. The mean across folds of the per-date Spearman rank IC was +0.0540, and the IC was positive on 65% of dates. When the date mean was subtracted from the target, mean R^2 became +0.0003, while the rank IC was unchanged to four decimals in every fold, as it must be because subtracting a per-date constant cannot reorder stocks. The date-mean component accounted for 28.6% of the pooled variance of the target over all cards from 2011 to 2026. These diagnostics carry no intervals and are descriptive. They show why the classification labels were defined as within-date ranks.

6 Limitations

1. **No project-level hold-out.** Every number uses 2014–2025 test years, and training windows and design choices drew on 2022–2025 (Section 2).
2. **Hindsight comparator.** The momentum-minus-volatility rule was chosen after the Phase 3a tilts were seen. Its 21.3% CAGR and 1.22 Sharpe are not evidence of a prospective strategy. Momentum [5] and low volatility [1] are long-established characteristics, which makes the rule plausible but does not remove the hindsight.

3. **Classifier intervals pending a block rerun.** The Phase 2 intervals use i.i.d. month resampling. The rule-1 verdicts for outperform and top performer rest on lower bounds of 0.0002 and 0.0006 and may not survive 2–6-month blocks.
4. **Volatility baseline is approximate.** The persistence comparison above is binned, pooled and post hoc; the exact per-date comparison is pending.
5. **Costs, capacity and exposures.** All returns are gross. The illustrative cost line is about 1.7%/yr for pick outperformers, about half its gross active return. Spreads, market impact and capacity of equal-weighted small caps were not modelled. Market beta and factor exposures were not estimated, so lower market exposure may contribute to every Sharpe gain reported.
6. **Multiple comparisons and selection.** The 15 `SBX` verdicts, and several unrecorded looks at the same 2014–2025 sample in earlier sandbox stages, mean that only results surviving the wider intervals should be treated as established. A deflated-Sharpe adjustment [2] was not computed.
7. **Specification.** The models are unregularised. Regularisation is outside the scope of this paper. The training objective (log-loss for an event across all stocks) differs from the portfolio's use of the scores (top-quintile ranking). This is one plausible, untested reason the fitted weights trailed the 1:1 rule. Others are estimation noise from annual three-year refits and weight placed on noisy features.
8. **Data.** April–June 2025 is missing, and the exchange-closed-date issue is unresolved (Section 2). The July–December 2018 universe may be misclassified.

What this paper does not establish. It does not show that any of these portfolios would be profitable after costs, that the momentum-minus-volatility rule would beat a fitted model on future data, or that the Sharpe gains are stock-selection alpha rather than lower market exposure. It does not show that the five features are uncorrelated on every date, or that fitted classifiers are generally inferior to simple rules beyond these five features, labels, universe and folds. Failure to detect a difference, as for avoid wild movers, is not evidence that no difference exists.

7 Conclusion

Across the tested rolling folds, the five-characteristic logistic models contained little information about which stocks would beat the cross-sectional median. Relative future volatility was substantially more predictable, largely because volatility persists. The avoid portfolios largely reproduced a simple exclusion of the most volatile fifth. Their incremental Sharpe advantage over that rule was small, marginal at 95% for the crash model, and not robust to the three-comparison interval. The outperform portfolio did not beat the momentum-minus-volatility comparator, which was designed with hindsight. The sample includes the project's previously reserved hold-out, the volatility feature has a documented missing-data episode, the portfolio results are gross of costs, and exposures were not estimated. The findings are therefore diagnostic, not prospective evidence of a tradable strategy. The next steps are:

- the specified moving-block rerun of the classifier intervals, with the exact current-volatility baseline;
- exposure regressions for the portfolios;
- a frozen-protocol forward test of the fitted model and the comparator.

Reproducibility

Use of AI tools. Code for this project was written with the assistance of an AI coding assistant (Anthropic's Claude), working to specifications set by the author, and the manuscript was drafted with the same assistance. The author designed the research questions and decision

Item	Setting
Dataset	`ml_sandbox_v1_dataset_2026-09-07`: 1,813,407 cards, 1,041 tickers, 3,756 dates, 2011-01-03 to 2026-06-10
Phase 2 (run 2026-09-30)	12 folds, train 3 y / test 1 y, 21-date purge; test cards 1,411,188 (high vol 1,411,156); scikit-learn `LogisticRegression` (no penalty, L-BFGS, max_iter 5000); i.i.d. calendar-month bootstrap, $B = 1000$, seed 0, 95% percentile
Phase 3a (2026-10-04)	141 rebalances (144 planned; Apr–Jun 2025 sat out), median 490 stocks (min 410); paired moving-block bootstrap, 6-month blocks, $B = 2000$, seed 0; 95% and 98.75%
Phase 3b (2026-10-05)	same months and bootstrap; 95% and 98.33%
Persistence check (post hoc)	decile rates from the conditional-probability notebook (trailing-252-day deciles, 2012-01-07 to 2026-06-10); equal-share binned AUC computed in this revision
Pending rerun	`rerun_phase2_bootstrap.py`: moving blocks of 2/3/6 months, $B = 1000$, seeds 20261016/20261017/20261020; exact current-volatility AUC (block 3, seed 314); per-date correlation and PCA distributions; log-turnover by year
Software	Python 3 on Google Colab; library versions were not recorded in the notebooks

rules and ran the main analyses. Some diagnostics added during revision were computed by the assistant from archived run outputs; these are labelled as revision diagnostics in the text. The author reviewed all code and text and takes full responsibility for the content.

Data and code availability. Code: https://github.com/mriganksahi-beep/indian-equity-research/tree/main/p4_models_vs_rules. The point-in-time panel construction is documented in https://github.com/mriganksahi-beep/indian-equity-research/tree/main/data_pipeline. Price data from Yahoo Finance and index-membership snapshots from a licensed vendor are not redistributed; the repository contains the scripts to rebuild the panel.

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